WORKED SOLUTION

STEP II 1992, Question 2

Pure • Clairaut Differential Equation

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q2

The key idea

The equation has the form $y = xp - f(p)$ with $p = \mathrm{d}y/\mathrm{d}x$, known as a Clairaut equation. Differentiating once makes the awkward terms p cancel, and what is left factorises into two much simpler equations. The catch is that differentiating loses information (constants disappear), so every candidate solution must be substituted back into $(*)$ to fix its constants.

Solution

Write $p = \dfrac{\mathrm{d}y}{\mathrm{d}x}$, so that $(*)$ reads $y = xp - \cosh p$.

Differentiating

Differentiate both sides with respect to x, using the product rule on xp and the chain rule on $\cosh p$:

$$p = p + x\frac{\mathrm{d}p}{\mathrm{d}x} - \sinh p\,\frac{\mathrm{d}p}{\mathrm{d}x}.$$

The p on each side cancels, leaving

$$\frac{\mathrm{d}p}{\mathrm{d}x}(x - \sinh p) = 0, \qquad \text{that is,} \qquad \frac{\mathrm{d}^2y}{\mathrm{d}x^2}\left(x - \sinh\frac{\mathrm{d}y}{\mathrm{d}x}\right) = 0.$$

A product is zero only if one of its factors is, so either $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$ or $x - \sinh\left(\dfrac{\mathrm{d}y}{\mathrm{d}x}\right) = 0$, as required.

The two equations

First equation. Integrating $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$ twice gives the general solution

$$y = Ax + B, \qquad A, B \text{ arbitrary constants.}$$

Second equation. Here $\sinh p = x$. Since sinh is a bijection from $\mathbb{R}$ to $\mathbb{R}$, this gives $p = \sinh^{-1} x$ for every x. Integrating by parts with $u = \sinh^{-1} x$, $\mathrm{d}v = \mathrm{d}x$:

$$\begin{aligned} y &= \int \sinh^{-1} x \,\mathrm{d}x = x \sinh^{-1} x - \int \frac{x}{\sqrt{1+x^2}} \,\mathrm{d}x \\ &= x \sinh^{-1} x - \sqrt{1+x^2} + C, \qquad C \text{ an arbitrary constant.} \end{aligned}$$

Solutions of $(*)$

Every solution of $(*)$ satisfies one of the two derived equations, but not conversely, because differentiating threw away a condition. So we substitute each family back into $(*)$.

> **Ask yourself** The derived equations have two and one arbitrary constants. Why can't all of these solve a first-order equation like $(*)$?

A first-order equation should have a one-parameter family of solutions, and each family above has too many constants or an unchecked one. Differentiating $(*)$ only tells us that $y - xp + \cosh p$ is *constant*; we need that constant to be zero.

Lines. With $y = Ax + B$ we have $p = A$, and $(*)$ becomes

$$Ax + B = Ax - \cosh A \iff B = -\cosh A.$$

So we get the one-parameter family of straight lines

$$y = Ax - \cosh A, \qquad A \in \mathbb{R}.$$

The other curve. With $y = x \sinh^{-1} x - \sqrt{1+x^2} + C$ we have $p = \sinh^{-1} x$, so $\sinh p = x$ and $\cosh p = \sqrt{1 + \sinh^2 p} = \sqrt{1+x^2}$ (taking the positive root because $\cosh p > 0$). The right-hand side of $(*)$ is then

$$xp - \cosh p = x \sinh^{-1} x - \sqrt{1+x^2},$$

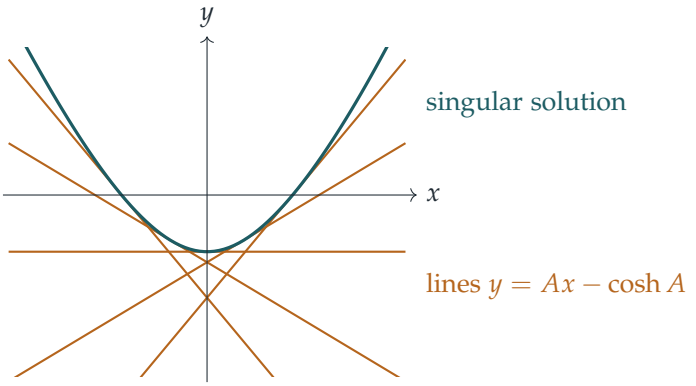
which equals y exactly when $C = 0$. This gives one extra solution, which contains no arbitrary constant:

$$y = x \sinh^{-1} x - \sqrt{1+x^2}.$$

So the solutions of $(*)$ are the lines $y = Ax - \cosh A$ (for each real A) together with the single curve $y = x \sinh^{-1} x - \sqrt{1+x^2}$.

How the two fit together

The line with slope A touches the curve at $x = \sinh A$. There the curve's gradient is $\sinh^{-1}(\sinh A) = A$, and its height is $A\sinh A - \cosh A$, the same as the line's. The curve has $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = \dfrac{1}{\sqrt{1+x^2}} > 0$, so it is convex and every one of these tangent lines lies below it. The curve is the *envelope* of the family of lines, and is called the *singular solution*.



This also explains a subtlety: since the condition $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}(x - \sinh p) = 0$ holds point by point, a solution may switch between the two types. For example, following the curve up to $x = \sinh A$ and then continuing along its tangent line there gives a function with a continuous derivative that still satisfies $(*)$ everywhere. Pieces like this are built from the solutions found above, and the expected answer is the family of lines plus the singular curve.

Checks

- At $A = 0$ the line is $y = -1$, and the curve has $y(0) = 0 - 1 = -1$ with gradient $\sinh^{-1} 0 = 0$: they touch at $(0, -1)$, as predicted.
- Numerically, with $x = 0.7$: $\sinh^{-1} 0.7 \approx 0.6527$, so the curve gives $y \approx 0.4569 - 1.2207 = -0.7638$, and $xp - \cosh p \approx 0.4569 - 1.2207$ agrees. Adding any $C \neq 0$ to y leaves a mismatch of exactly C.
- Checking $(*)$ by finite differences at random points gives residuals at rounding-error level for every line $y = Ax - \cosh A$ and for the singular curve.

Common slips

- Stopping at $y = Ax + B$ or at $y = x\sinh^{-1} x - \sqrt{1+x^2} + C$: these solve the *differentiated* equation, and $(*)$ itself forces $B = -\cosh A$ and $C = 0$.

► Missing the singular solution altogether by "cancelling" the factor $\dfrac{\mathrm{d}p}{\mathrm{d}x}$.

Beyond STEP

Equations of the form $y = xp - f(p)$ are Clairaut equations. Their general solution is always the family of lines $y = cx - f(c)$, and the singular solution is the envelope of that family. Here $f = \cosh$, and the envelope $y = x\sinh^{-1}x - \sqrt{1+x^2}$ is the Legendre transform of cosh, a construction that appears in mechanics (passing from the Lagrangian to the Hamiltonian) and in thermodynamics. The example also shows that solutions of first-order equations need not be unique when the conditions of the existence and uniqueness theorem fail. Boyce and DiPrima, *Elementary Differential Equations*, covers first-order equations and that uniqueness theorem.

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